

each coloured row meets the input and gives
the sum of the node with the same colour

$$\begin{bmatrix} 0.2 & 0.4 & -0.5 & 0.3 \\ -0.3 & 0.1 & 0.2 & -0.4 \\ 0.5 & -0.2 & 0.1 & 0.2 \end{bmatrix} \begin{bmatrix} 0.72 \\ 0.72 \\ 0.69 \\ 0.81 \end{bmatrix} + \begin{bmatrix} 0.1 \\ -0.1 \\ 0.2 \end{bmatrix} = \begin{bmatrix} 0.430 \\ -0.430 \\ 0.647 \end{bmatrix}$$

$W^{1\top}$: 3×4 , one row per node

a^0 : 4×1

b^1

z^1 : 3×1